FLEET TEAM DIGEST

737MAX-FTD-28-18001

Issue Title : Fuel Pump Low Pressure Switch

Airplane Model	ATA	Minor Model(s)
737MAX	2822-00	-7, -8, -9

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/08/2018	06/08/2021	05/25/2021

Estimated Completion Date	Next Update Date

Revision Description

This revision includes information related to risks of protective boot installations on Center Tank pump switches. This Fleet Team Digest is now considered closed.

Applicability

All 737 MAX boost pump pressure switches.

Description

The outlet of each of the six fuel pumps used in the fuel system is equipped with a low pressure switch. These switches provide signals to the flight deck indicating when the fuel pump is not supplying fuel pressure at the pump outlet. Boeing received reports from several different 737NG operators that the fuel pressure switches were not operating properly. Many of the reports of poor performance were corrected by cleaning the electrical connector contacts. Evaluation of all other events and subsequent troubleshooting often find no fault with the fuel pump or the associated wiring; in these cases it was concluded that the pressure switch contributed to the erroneous failure message.

The vast majority of 737NG pressure switch failures were Aft Boost Pump pressure switches, which are located on the rear wing spar. Boeing believes that the tires are throwing water and debris almost directly on the switch. On 737 MAX, the Forward (Fwd) Boost Pump pressure switches have been relocated and are now alongside the Aft Boost Pump pressure switches on the rear wing spar. The pressure switches located on the rear wing spar are more susceptible to microswitch contamination and subject to higher failures when compared to the Center Tank Boost Pump switches installed on the rear spar within the wheel well.

Status

Boeing obtained several switches that were returned from service for confirmed faults and were subjected to a Boeing-internal analysis. The analysis concluded that some contaminants had been able to reach the microswitch inside the P/N 21C1-3-21 low

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pressure switch. Analysis showed the contaminants were likely potassium salts. It appears the contaminants entered through four weep holes that are drilled in the casing. The contaminants had coated the internal microswitch, preventing the microswitch from activating properly.

In 2010, Boeing intended to certify the P/N 21C1-6-7 hermetically sealed pressure switch as an optional spare part to P/N 21C1-3-21. It was not clear at the time if this new switch would have improved reliability because contaminants could still enter the weep holes. The spares cost of the P/N 21C1-6-7 is about twice as expensive as the current switch. Based on the cost of the new part and the unknown gain in reliability, Boeing cancelled this design effort.

As an alternate to the hermetically sealed pressure switch, Boeing worked with Eaton on this issue and implemented P/N JG203, which is a protective device, or "Boot," that will inhibit moisture from migrating into the switch body.

Boeing published Service Bulletin (SB) 737-28-1370 for 737 MAX airplanes, which provides a retrofit solution to 737 MAX airplanes placed in a delivery state prior to 18-November-2020. This SB will enable:

- sealing boot installation on Left/Right main tank pump pressure switches, P/N 21C1-3-21, and
- marker installation in the wheel well to prohibit the sealing boots on the Center tank pump pressure switches, P/N 21C1-3-22.

In addition, 737 MAX airplanes placed in a delivery state after 18-November-2020 will have the sealing boots and markers installed.

Boeing is aware that some operators have installed a Part Manufacturer Approval (PMA) part, which includes the sealing boot, at the Center Tank Boost Pump pressure switch location. Although the P/N JG203 sealing boot will fit both the 21C1-3-21 and 21C1-3-22 pressure switches, Boeing strongly advises against the installation of such boot on 21C1-3-22 switches. An incorrect installation of sealing boots on the Center Tank pressure switches, such as the vent hole up rather than vent hole down, could lead to water being trapped in the pressure switch and freeze during flight. The frozen water would keep the pressure switch in the high pressure position even after the Center Tank fuel has been used, and the pumps have gone low in output pressure. This would result in the Center Tank pumps running dry for the remainder of the flight since the automatic pump shutoff systems are dependent on the low pressure indications, and the flight crew may not be inclined to manually turn off the pumps due to a lack of low pressure indications. Analysis performed on other airplane models with similar pumps indicates that, under certain circumstances, pumps running dry for an extended period can lead to an ignition source in the tank. As regulations require that manufacturing variability be considered as part of the certification of the fuel system, and the indication design must minimize errors that could create hazards, it was decided that the pressure switch boot should not be installed on the Center Tank pressure switches.

Note: There is no intention to require that the pressure switch be replaced upon installation of the new “boot.” However, a previously installed pressure switch may already have some degree of degradation or moisture ingress. Should a new “boot” be

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installed over a previously installed pressure switch, subsequent failures should not necessarily be interpreted as a failure of the “boot” concept.

Interim Action

No interim actions have been defined for operators. If a low fuel pressure indication is experienced, operator maintenance should initially clean the electrical connectors and determine if that corrects the anomaly. If not, then switch replacement should be considered.

Final Action

737 MAX operators are encouraged to incorporate SB 737-28-1370.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Production / Delivery Incorporation: November 2020 (Complete)

Retrofit SB 737-28-1370 Available: 31-March-2021 (Complete)

Parts Available: (Complete)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Economic (FTE)	737NG-FTE-28-X2821		
Fleet Team Digest Article (FTD)	737NG-FTD-28-05004		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Pressure Switch Sealing Boot	N/A	EATON	JG203
Fuel Boost Pump Low Pressure Switch (Current)	60B92400-10	EATON	21C1-3-21
Fuel Override Pump Low Pressure Switch (Current)	60B92400-11	EATON	21C1-3-22

Related Categories

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- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- In Service Occurrences : null

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